

Sizhuo Qi

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TECHNICAL SKILLS

Programming Languages	C, C++, Python, SystemVerilog, Verilog, VHDL, Java, JavaScript, TypeScript, SQL (PostgreSQL)
Digital Design / FPGA	Quartus Prime, ModelSim, SignalTap, DE1-SoC (Cyclone V), DE0-CV (Cyclone V), PicoBlaze
Embedded / Firmware	STM32F103 (ARM Cortex-M3), GPIO, timers/PWM, ADC, DMA, I2C, UART, SPI, interrupt-driven control, Keil uVision, ST-LINK
Developer Tools	Git, VS Code, Linux/Unix, GDB, Jenkins, DBeaver
Testing / Debugging	unit testing, regression testing, TDD, Jest, JUnit
Circuit Design / Simulation	oscilloscopes, LTspice, Logisim, Antares, Multisim
Web / Frontend	HTML, CSS, Liquid, React.js, Node.js, Koa.js, Bootstrap

WORK EXPERIENCE

Service Router Platform Test Developer Co-op, Nokia	Sep 2026 — Present
- Worked in Linux-based IP routing test environments, using command-line tools to connect from development VMs to remote testbeds and access devices under test (DUTs). - Used Python-based pyshl tooling to execute automated regression tests on remote testbeds and interacted with SR Linux and SR OS DUTs through router CLIs to inspect system state and validate connectivity. - Developing automated validation for DUT-to-DUT links across test topologies, verifying that when both endpoint ports are enabled, both report operational state up. - Ramping up within the Sync & Timing team on PTP (IEEE 1588), GNSS, Tcl-based tooling, and timing-platform regression workflows.	
Full Stack Developer Co-op, BC Liquor Distribution Branch	Jan 2026 — Aug 2026
- Migrated legacy SOAP-based enterprise applications from Java 8 / WildFly 14 to Java 17 / WildFly 33 (Jakarta EE 10), resolving deployment, dependency, and compatibility issues. - Deployed EAR applications, updated Maven/WSDL dependencies, supported Jenkins CI/CD workflows, and performed integration, regression, and unit testing using SoapUI and Maven.	
Web Developer full-time, Coast Appliances Company	Feb 2022 — Aug 2024
Developed full-stack web applications and backend integrations using JavaScript, React.js, Node.js, REST APIs, SQL, and GraphQL, supporting payment, product, order, and data synchronization workflows.	

PROJECTS

Light & Thermistor Monitoring System using STM32F103 (Personal Project)	Sep 2026
- Designed and implemented a dual-sensor embedded monitoring system on STM32F103 using C and STM32 HAL, integrating an analog light sensor and NTC thermistor for continuous environmental sensing. - Configured TIM3 TRGO, ADC1 multi-channel scan mode, and circular DMA to sample both sensor channels at 100 Hz without CPU polling. - Implemented 8-sample moving-average filtering, TIM2 Channel 3 PWM for proportional LED brightness control, and hysteresis-based temperature warning logic to prevent output chatter near threshold boundaries. - Added USART1 telemetry for real-time light and temperature diagnostics, and documented the hardware configuration and firmware in Git/GitHub (GitHub repository)	
FPGA Song Player on DE1-SoC	May 2025 — Jun 2025
- Developed a synthesizable music player in SystemVerilog on the DE1-SoC FPGA, implementing playback controls, clock division, and synchronized audio data generation. - Designed FSM-based control and memory-address sequencing logic, and verified functional behavior and timing using ModelSim testbenches and SignalTap on-chip debugging. - Synthesized and deployed the design using Quartus, integrating a PicoBlaze soft processor for average-signal computation and LED display.	

EDUCATION

Bachelor of Computer Science GPA: A+ (90.8%) Dean's Scholar University of British Columbia	Expected 2028/2029
Master of Electrical and Computer Engineering GPA: A (86%) University of British Columbia	Sep 2019 — Jan 2022
Bachelor of Engineering GPA: A (89%) Nanjing University of Aeronautics and Astronautics	Sep 2013 — Jun 2017